

# Danial Kamali

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## Education

### Michigan State University

PhD in Computer Science (GPA: 4.0, **Research:** Vision & Language)

East Lansing, Michigan, USA

Aug 2022 - Present

### Iran University of Science and Technology

B.Sc in Computer Engineering (**Research:** Facial Expression Recognition, NLP Lexical Analysis)

Tehran, Iran

Sep 2016 - March 2021

## Experiences

### Michigan State University

Graduate Research Assistant

East Lansing, USA

Sep 2022 - Present

- Implemented various deep learning tasks in DomiKnowS (A Library for Integration of Symbolic Domain Knowledge in Deep Learning) for the paper.
- Researching Compositional Generalization on multi-modal systems improves generalization in unseen structures.
- Lead a team of researchers on data gathering, labeling, and baseline preparation on the misinformation detection task.

### University of California, Los Angeles

Visiting Graduate Student

Los Angeles, USA

Sep 2025 - Dec 2025

- Conducting research on vision-language models with a focus on compositional reasoning.

### Qualtrics<sup>XM</sup>

Applied Science Intern

Seattle, USA

Jun 2025 - Aug 2025

- Conducted AI/ML research at the Data Intelligence Center of Excellence (DICE).
- Revamped the Qualtrics Assist data analytics agent, leading to a 20% improvement in performance.
- Boosted recall in context retrieval through advanced retrieval and context reduction strategies.
- Enabled the system to handle a broader range of diverse and compositional question types efficiently.

### Zibal

Backend Team Leader/Full-Stack Developer

Tehran, Iran

Sep 2021 - Aug 2022

- Stack:** ReactJS, Ant Design, LESS.
- Managed a team of 4 people with the Agile manifesto and improved the team capacity utilization.
- Reviewed backend team merge requests on Gitlab and reduced the defection rate.
- Mentored two interns through 8 weeks of training, one of them joined our team as a junior developer.

Backend Developer

Oct 2019 - Aug 2021

- Stack:** Django, DRF, TensorFlow2, MongoDB, Postgres, Redis, Celery, Docker, Kubernetes
- Developed data aggregation pipelines and dashboard generation for BI system from NOSQL to SQL-based DB for the product team.
- Implemented Django-MongoDB generic view for auto dashboard design generation, leveraging reusable patterns to increase developer productivity.
- Designed CI/CD using Gitlab and kubernetes for front-end and back-end and automated the development and deploy operations.

### IUST Data Mining Lab

Natural Language Processing Research Assistant

Tehran, Iran

Jun 2019 - Dec 2021

- Evaluated Farsi Sentences Tokenization and analyzed current Farsi tokenizers.
- Mentored undergrad thesis projects and provided feedback on different stages of their project.

### IUST Multi-Agent Systems Lab

Machine Learning Research Assistant

Tehran, Iran

Jun 2019 - Sept 2019

- Researched on Image Caption Generation and customized image captioning in Farsi.
- Gathered the Farsi image captioning dataset using the coco dataset.
- Trained the model using the attention mechanism on the mentioned dataset, using TensorFlow.

## Selected Publications

- Kamali, Danial**, & Kordjamshidi, P. "NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language." *International Conference on Learning Representations (ICLR)*, 2026.
- Nafar, A., **Kamali, Danial**, Karimian, H. R., & Kordjamshidi, P. "An Agentic Framework for Neuro-Symbolic Programming." EACL 2026. Under Review.
- Sinha, S., Premisri, T., **Kamali, Danial**, & Kordjamshidi, P. "Neuro-Symbolic Frameworks: Conceptual Characterization and Empirical Comparative Analysis.", 2025. Under Submission.
- Kamali, Danial**, Barezi, E. J., & Kordjamshidi, P. "NeSyCoCo: A Neuro-Symbolic Concept Composer for Compositional Generalization." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- Kamali, Danial**, Romain, J., Liu, H., Peng, M., & Kordjamshidi, P. "Misinformation Detection Using Persuasive Writing Strategies." *Joint International Conference on Computational Linguistics, Language Resources, and Evaluation (COLING)*, 2024.
- Kamali, Danial**, & Kordjamshidi, P. "Syntax-Guided Transformers: Elevating Compositional Generalization and Grounding in Multimodal Environments." *GenBench: Workshop on Generalization Benchmarking in NLP, Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

## Skills

### Programming

Python, Java Script, C/C++, JAVA, C#

### AI Frameworks

Sklearn, Tensorflow, PyTorch, Keras, HuggingFace, Numpy, OpenCV, NLTK, LangGraph, Langchain

### Tools

Kubernetes, Docker, MySQL, MongoDB, Redis, Celery, Kafka

### Frameworks and Libraries

Django, DRF, React, Express, Flask, Unity

### Language

Farsi: Native

English:

Professional

Projects

Neural Machine Translation

- Initiated the development of Finglish ol'Saltalneh, an NMT model for Farsi-to-English transliteration, enhancing translation for English forms.
- Curated a dataset of 100,000+ Farsi-English samples, leading data-cleaning efforts to standardize input and improve model training outcomes.
- Devised and applied data augmentation techniques to bolster the dataset, resulting in a 20% BLEU score enhancement for the model.

Facial Expression Recognition

- Developed a convolutional neural network (CNN) that achieved state-of-the-art accuracy on the RAF-DB dataset and competitive results on FER2013.
- Implemented data augmentation and transfer learning techniques to improve model generalizability.
- Implemented statistically aware layers to the network, increasing the interpretability of facial feature representations.

Memotion Analysis: MultiModal Image-Text classification

- Engaged in Memotion Analysis competition on CodaLab, developing a multimodal classification system using NLP and Computer Vision.

Real-time Augmented Reality Sudoku Solver

- Engineered a real-time AR application that captures Sudoku puzzles via video stream, solves them, and overlays the solutions.
- Implemented a custom digit recognition algorithm using neural networks that achieved 95% accuracy on handwritten digits.
- Developed the AR overlay system using OpenCV and Unity, resulting in an interactive and user-friendly application experience.

Awards

|            |                                                                                                |                   |
|------------|------------------------------------------------------------------------------------------------|-------------------|
| 2025       | <b>Recipient</b> COGS Professional Development award                                           | MSU               |
| 2025       | <b>Recipient</b> Graduate School Research Enhancement Award                                    | MSU               |
| 2022       | <b>Recipient</b> Dr. Carl Page Sr Fellowship                                                   | MSU               |
| 2020       | <b>Ranked 1st</b> Annual evaluation of CE department students                                  | IUST              |
| 2015       | <b>Ranked among the top 0.2% of the candidates</b> National Entrance Exam for Graduate Schools | National          |
| 2013, 2014 | <b>Qualified for the second phase</b> National Mathematics Olympiad                            | National          |
| 2013       | <b>Qualified for the Iran National Team</b> International Mathematics Competition              | National          |
| 2012       | <b>Exceptional Talent Scholarship</b> Entrance Exam                                            | Salam High School |

Talks

|      |                                                                                                                    |            |
|------|--------------------------------------------------------------------------------------------------------------------|------------|
| 2025 | <b>Compositional Reasoning Under Perceptual Uncertainty</b> UCLA NLP                                               | LA         |
| 2025 | <b>NeSyCoCo: A Neuro-Symbolic Concept Composer for Compositional Generalization</b> Midwest Speech & Language      | Notre Dame |
| 2025 | <b>NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language</b> ResearchTrend.AI | Germany    |
| 2025 | <b>NePTune: A Neuro-Pythonic Framework for Tunable Compositional Reasoning on Vision-Language</b> SpaVLE NeurIPS   | San Diego  |

Services & Activities

|           |                                                     |                   |
|-----------|-----------------------------------------------------|-------------------|
| 2024      | <b>Student Volunteer</b> EMNLP'24 GenBench Workshop | MSU               |
| 2022-2025 | <b>Reviewer</b> ACL'23, EMNLP'23, IJACI'24, AAAI'26 | MSU               |
| 2023      | <b>President</b> Iranian Students Association       | MSU               |
| 2022      | <b>Treasurer</b> Iranian Students Association       | MSU               |
| 2022      | <b>Development Team</b> GAMEIN Event                | Sharif University |

Selected Academic Courses

|                                                                                                                                                                                                                                      |                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MSU                                                                                                                                                                                                                                  | IUST                                                                                                                                                                                     |
| <ul style="list-style-type: none"><li>• Deep Learning: 4.0 (<b>top 3</b>)</li><li>• Computational Foundation of AI: 4.0 (<b>top 1</b>)</li><li>• Natural Language Processing: 4.0</li><li>• Advanced Algorithm Design: 4.0</li></ul> | <ul style="list-style-type: none"><li>• Computer Vision: 4.0</li><li>• Deep Learning: 4.0</li><li>• Computational Intelligence: 4.0</li><li>• Natural Language Processing: 4.0</li></ul> |